

Fourier Series

Ryan McLean

June 2026

1 Introduction

Fourier Analysis is weird, but lets get in to it. We are going to focus on Fourier series. The basic idea of Fourier series is any repeating function can be replaced with a sum of sinusoidal functions. (Don't use the word cosinusoidal, its redundant and unnecessary see definitions as to why.) The process is similar to Taylor series. Just as a Taylor series represent a continuously differentiable function from its derivatives, the Fourier transform represents a periodic function from sine and cosine waves.

Thus any repetitive (periodic) function has a Fourier series representation. This is powerful when we restrict the domain of a function and define it as a piecewise repeating function, with some extra qualifications we can obtain a Fourier series over the limited domain! Isn't that exciting! I know your not excited, but do try. Building on this, when we model systems using partial differential equations finding solutions to partial differential equations is hard. But fortunately finding Fourier series representations of solutions is somewhat manageable, so we often uses Fourier series in finding solutions to models applying partial differential equations.

Another example of Fourier series usefulness is signal analysis. This is one of its primary uses in engineering, however, it is widely applicable elsewhere. A noticeable amount of communications is done with periodic models. So what do we use to simplify or solve our math? Fourier series!

2 Definitions and Theorems

Sinusoidal Function

A sinusoidal function is a function of the form $f : \mathbb{R} \rightarrow \mathbb{R}$ with

$$f(x) = A \sin(\omega x + \phi) + D$$

where $A, \omega, \phi, D \in \mathbb{R}$

Cosinusoidal Function

A cosinusoidal function is a function of the form $f : \mathbb{R} \rightarrow \mathbb{R}$ with

$$f(x) = A \cos(\omega x + \phi) + D$$

where $A, \omega, \phi, D \in \mathbb{R}$

Observe and recall that $\sin(x) = \cos(x - \frac{\pi}{2})$ so sinusoidal and cosinusoidal are the same thing.

Periodic

Let $P \in \mathbb{R}$, and $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function. We say f is periodic with period P if

$$f(x + P) = f(x)$$

$\forall x \in \mathbb{R}$

Fourier Series

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ has a Fourier series representation, and let $n \in \mathbb{N}$. Then f is periodic, and suppose that period is $2P$. Then

$$f(x) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(nx) + b_n \sin(nx)]$$

With the coefficients a_0 , a_n , and b_n given by

$$a_0 = \frac{1}{2P} \int_{-P}^P f(x) dx$$

$$a_n = \frac{1}{P} \int_{-P}^P f(x) \cos(nx) dx$$

$$b_n = \frac{1}{P} \int_{-P}^P f(x) \sin(nx) dx$$

Fourier Transform

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function. There are other restrictions to assume. Then the Fourier transform is defined as

$$\hat{f}(s) = \int_{-\infty}^{\infty} f(x) e^{-i2\pi xs} dx$$

3 Examples

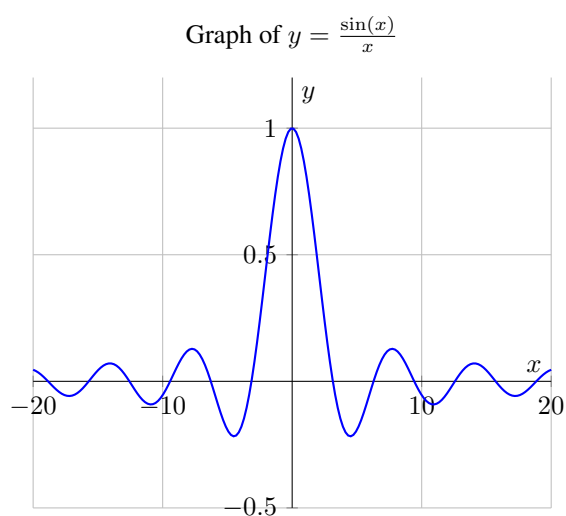
Lets start our examples by classifying some periodic and non periodic functions. Consider the following examples:

$$f(x) = \frac{\sin(x)}{x}$$

$$g(x) = \tan(x) \sec(x) + \cos(x)$$

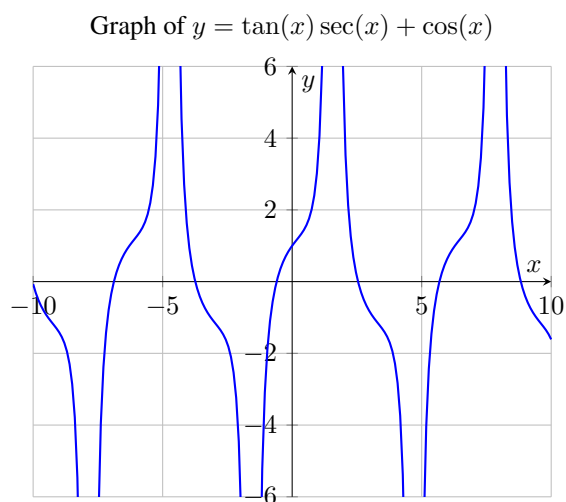
$$h(x) = e^x$$

Lets start by looking at this problem graphically. Starting with f :



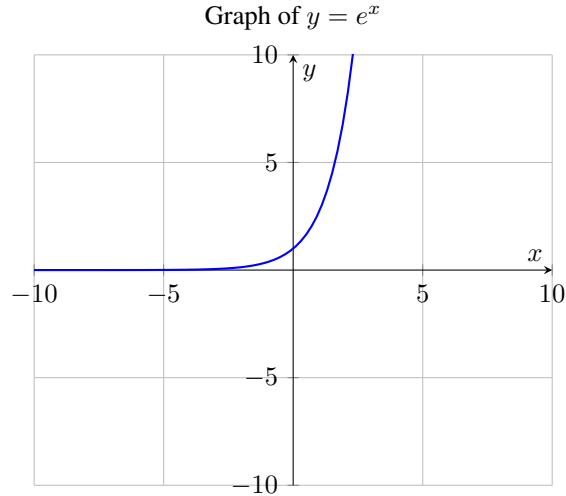
We can see that although there is a clear pattern there is no repeating pattern making this not periodic.

Lets now look at g :



As we can see there is a repeating pattern for $g(x) = \tan(x) \sec(x) + \cos(x)$ thus we have a periodic function. Further g has period 2π .

Finally looking at h we have the graph:



And seen graphically or from previous knowledge we know $h(x) = e^x$ is a one-to-one function which means it cannot be periodic as there are no repeated output values of any kind let alone a repeating consistent pattern.

$$\text{f} \quad f(x) = \frac{\sin(x)}{x} \quad \text{not periodic}$$

$$\text{So we have: g} \quad g(x) = \tan(x) \sec(x) + \cos(x) \quad \text{periodic}$$

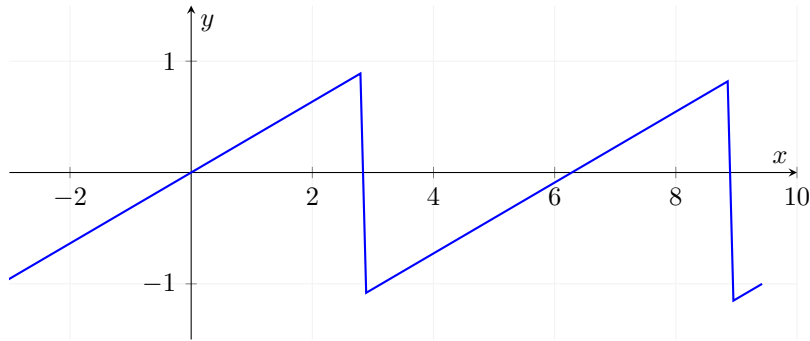
$$\text{h} \quad h(x) = e^x \quad \text{not periodic}$$

At the end we will return to algebraically demonstrate f is not periodic, but to get a good intuition I recommend playing with some sort of graphing calculator.

Lets now perform some Fourier analysis and build a Fourier series. Lets look at the classic example of the saw tooth function. The saw tooth function is of the form

$$f(x) = \begin{cases} \frac{1}{\pi}x & \text{if } -\pi < x \leq \pi \\ f(x + 2\pi k) & \text{otherwise where } k \text{ sacles into } -\pi < x \leq \pi \end{cases}$$

The saw tooth function then has the graph:



Lets now build the Fourier series on this function. To start lets calculate a_0 . First observe the period is 2π , so we will plug in $P = \pi$ as recall $2P$ is the periode in our formula.

$$a_0 = \frac{1}{2P} \int_{-P}^P f(x) dx$$

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx$$

And as we are only integrating over one interval of the peace wise function we will substitute $f(x) = \frac{x}{\pi}$.

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{x}{\pi} dx$$

$$= \frac{1}{2\pi^2} \int_{-\pi}^{\pi} x dx$$

$$= \frac{1}{2\pi^2} \left(\frac{x^2}{2} \right) \Big|_{-\pi}^{\pi}$$

$$= \frac{1}{4\pi^2} (\pi^2 - (-\pi)^2)$$

$$a_0 = 0$$

So we have calculated $a_0 = 0$, lets now look at a_n , again recalling $P = \pi$.

$$a_n = \frac{1}{P} \int_{-P}^P f(x) \cos(nx) dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx$$

Plugging in our function

$$\begin{aligned}
a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} \frac{x}{\pi} \cos(nx) dx \\
&= \frac{1}{\pi^2} \int_{-\pi}^{\pi} x \cos(nx) dx
\end{aligned}$$

Applying integration by parts which is

$$\int v du = uv - \int u dv$$

So lets build a table

$$u = x \quad dv = \cos(nx) dx$$

$$du = 1 dx \quad v = \frac{\sin(nx)}{n}$$

This yields

$$\begin{aligned}
a_n &= \frac{1}{\pi^2} \int_{-\pi}^{\pi} x \cos(nx) dx \\
&= \frac{1}{\pi^2} \left(x \frac{\sin(nx)}{n} - \int 1 \frac{\sin(nx)}{n} dx \right) \Big|_{-\pi}^{\pi} \\
&= \frac{1}{\pi^2} \left(x \frac{\sin(nx)}{n} - \frac{\cos(nx)}{n^2} \right) \Big|_{-\pi}^{\pi}
\end{aligned}$$

Recall even and odd functions from the trig section in the identity's notes document. And the trig identity for even odd functions giving $\cos(-x) = \cos(x)$ as cosine is even and $\sin(-x) = -\sin(x)$ as sine is an odd function. This simplifies our expression as the $\frac{\cos(nx)}{n^2}$ term goes to zero under the evaluation.

$$\begin{aligned}
a_n &= \frac{1}{\pi^2} x \frac{\sin(nx)}{n} \Big|_{-\pi}^{\pi} \\
&= \frac{1}{\pi^2} \left(\pi \frac{\sin(n\pi)}{n} - (-\pi) \frac{\sin(n(-\pi))}{n} \right) \\
&= \frac{1}{\pi^2} \left(\pi \frac{\sin(n\pi)}{n} - \pi \frac{\sin(n\pi)}{n} \right) \\
&= 0
\end{aligned}$$

So we have that $a_n = 0, \forall n \in \mathbb{N}$

Finally lets calculate b_n .

$$\begin{aligned}
 b_n &= \frac{1}{P} \int_{-P}^P f(x) \sin(nx) dx \\
 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx
 \end{aligned}$$

Plugging in $f(x) = \frac{x}{\pi}$

$$\begin{aligned}
 b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} \frac{x}{\pi} \sin(nx) dx \\
 &= \frac{1}{\pi^2} \int_{-\pi}^{\pi} x \sin(nx) dx
 \end{aligned}$$

Applying integration by parts

$$u = x \quad dv = \sin(nx) dx$$

$$du = 1 dx \quad v = \frac{-\cos(nx)}{n}$$

yeilding

$$\begin{aligned}
 b_n &= \frac{1}{\pi^2} \left(x \left(\frac{-\cos(nx)}{n} \right) - \int 1 \frac{-\cos(nx)}{n} dx \right) \Big|_{-\pi}^{\pi} \\
 &= \frac{1}{\pi^2} \left(-x \left(\frac{\cos(nx)}{n} \right) + \int \frac{\cos(nx)}{n} dx \right) \Big|_{-\pi}^{\pi} \\
 &= \frac{1}{\pi^2} \left(-x \left(\frac{\cos(nx)}{n} \right) + \frac{\sin(nx)}{n^2} \right) \Big|_{-\pi}^{\pi}
 \end{aligned}$$

Applying even odd identity's again

$$b_n = \frac{1}{\pi^2} 2 \frac{\sin(n\pi)}{n^2}$$

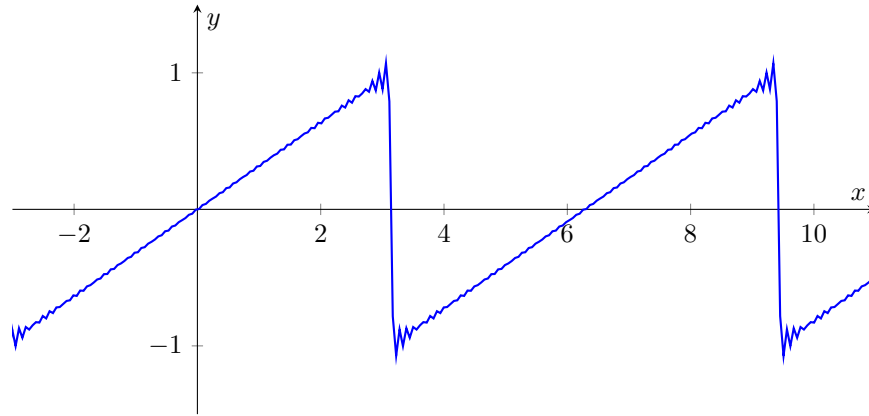
so we have

$$b_n = \frac{2 \sin(n\pi)}{(n\pi)^2}$$

This then gives the Fourier series

$$\begin{aligned} f(x) &= a_0 + \sum_{n=1}^{\infty} [a_n \cos(nx) + b_n \sin(nx)] \\ &= \sum_{n=1}^{\infty} \frac{2 \sin(n\pi)}{(n\pi)^2} \sin(nx) \end{aligned}$$

Turning to look at the graph of the first 50 terms we have the figure:



To end let's return to periodic functions and look at $f = \frac{\sin(x)}{x}$. Is there some P such that $f(x + P) = f(x)$? To see let's suppose $\exists P \in \mathbb{R}$ such that $f(x + P) = f(x)$. Then we have

$$f(x + P) = f(x)$$

$$\frac{\sin(x + P)}{x + P} = \frac{\sin(x)}{x}$$

So

$$\sin(x + P)x = \sin(x)(x + P)$$

$$\sin(x + P)x = \sin(x)x + \sin(x)P$$

$$\sin(x + P)x - \sin(x)x = \sin(x)P$$

$$(\sin(x + P) - \sin(x))x = \sin(x)P$$

$$\frac{\sin(x + P) - \sin(x)}{\sin(x)} = \frac{P}{x}$$

$$\frac{\sin(x + P)}{\sin(x)} - \frac{\sin(x)}{\sin(x)} = \frac{P}{x}$$

$$\frac{\sin(x + P)}{\sin(x)} - 1 = \frac{P}{x}$$

recall the sum identity's of the sine and cosine functions.

$$\sin(x \pm y) = \sin(x) \cos(y) \pm \cos(x) \sin(y)$$

$$\cos(x \pm y) = \cos(x) \cos(y) \mp \sin(x) \sin(y)$$

Which yeilds

$$\frac{\sin(x) \cos(P) + \cos(x) \sin(P)}{\sin(x)} - 1 = \frac{P}{x}$$

And note since P is constant so to are $\sin(P)$ and $\cos(P)$.

$$\cos(P) + \frac{\cos(x) \sin(P)}{\sin(x)} - 1 = \frac{P}{x}$$

$$\cos(P) + \cot(x) \sin(P) - 1 = \frac{P}{x}$$

$$x = \frac{P}{\cos(P) + \cot(x) \sin(P) - 1}$$

And since this must be true $\forall x$ if the original f is periodic we have an implication that x is a nonlinear function of itself which is a contradiction. Thus we can conclude $f(x) = \frac{\sin(x)}{x}$ is not periodic.